

Getting started with Nmap and port scanning

Nmap is a versatile tool available on multiple operating systems, including Kali Linux, macOS, and other Linux distributions. Given below are the steps to get started on each platform.

Kali Linux: Kali Linux, a Debian-based distribution, is specifically designed for penetration testing and comes with Nmap pre-installed. To verify its installation, type:

```
nmap --version
```

If Nmap is not installed, you can install it using:

```
sudo apt-get update
sudo apt-get install nmap
```

macOS: On macOS, Nmap can be installed using package managers like Homebrew. First, ensure Homebrew is installed, and then execute:

```
brew install nmap
```

After installation, verify by checking the version:

```
nmap --version
```

Other Linux distributions: For distributions like Ubuntu or Fedora, Nmap can be installed using the respective package managers. For Ubuntu, use:

```
sudo apt-get update
sudo apt-get install nmap
```

For Fedora, use:

```
sudo dnf install nmap
```

Once installed, Nmap can perform various types of scans.

Ping scan: To discover active hosts on a network, type:

```
nmap -sn 192.168.1.0/24
```

Port scan: To identify open ports on a specific host, use:

```
nmap 192.168.1.1
```

Service version detection: To determine the version of services running on open ports, type:

```
nmap -sV 192.168.1.1
```

Operating system detection: To identify the operating system of a host, use:

```
nmap -O 192.168.1.1
```

Intrusion detection and continuous monitoring with Nmap

Nmap plays a pivotal role in intrusion detection and continuous monitoring, empowering security professionals to proactively identify anomalies and potential security breaches. As cyber threats become more sophisticated, organisations require robust network security tools to detect vulnerabilities and mitigate risks in real-time. Nmap's comprehensive scanning capabilities make it an essential component of network defence strategies, providing visibility into network activity and identifying unauthorised access attempts before they escalate into major security incidents.

One of the primary benefits of Nmap in intrusion detection is its ability to scan networks for open ports, running services, and device fingerprints. Security professionals use these scans to establish a baseline of normal network behaviour. Any deviations from this baseline—such as the sudden appearance of unauthorised services, unexpected open ports, or changes in device configurations—could indicate a potential cyberattack, malware infection, or unauthorised access. Nmap's ability to conduct real-time, on-demand, and scheduled scans ensures that even subtle changes in network infrastructure are detected promptly.

Nmap also facilitates continuous monitoring by integrating with security information and event management (SIEM) systems. By automating network scans and feeding real-time data into SIEM platforms, security teams can correlate network events, detect intrusion attempts, and respond to security threats in a timely manner. This integration enhances an organisation's threat intelligence by providing a centralised view of network activity, reducing false positives, and prioritising security alerts.

Beyond intrusion detection, Nmap is a valuable tool for incident response and forensics analysis. When a security breach occurs, investigators use Nmap to analyse compromised systems, identify backdoors, and assess the full extent of an attack. The tool helps in mapping attacker footprints, uncovering suspicious IP addresses, and detecting hidden services that may have been exploited by cybercriminals. Its ability to detect host-based intrusion indicators, such as unauthorised SSH access or abnormal traffic patterns, enables security teams to contain threats and remediate vulnerabilities quickly. Additionally, Nmap supports custom scripting through the Nmap Scripting Engine (NSE), which allows for automated vulnerability detection, malware scanning, and compliance auditing. Security analysts